MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 12: Non-Parametric Structural Equations and Directed Acyclic Graphs

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Readings: *Applied Causal Inference Powered by ML and AI*, §7, §11.

Where We Are

We have completed a long journey through causal machine learning estimation:

- **Potential outcomes framework:** $Y(1), Y(0)$, ATE, ATT, CATE.
- **Key identification assumption:** Conditional exogeneity $Y(d) \perp\!\!\!\perp D \mid X$.
- **Two identification strategies:** Conditioning & propensities.
- **Estimation paradigms:** DR/DML estimators; DR-Learner and R-Learner for CATE.
- **Overarching principle:** "Insensitive estimation formulas" robust to nuisance model errors.

We emerged victorious from causal ML estimation! Now we switch gears completely.

The Big Question

We go back to the basics — the real "Causal Inference."

The Motivating Question

The main assumption that enabled identification was **conditional exogeneity**. But how can we *deduce* it from more **human-interpretable** assumptions?

Imagine going to a domain expert and asking:

> *"Do you believe that your potential outcomes are conditionally independent of the treatment given the covariates?"*

Most domain experts won't even grasp what this means!

Our goal: Find a more **human-friendly** way of expressing assumptions on the data generating process, such that conditional exogeneity is a **conclusion** — not the premise.

What We Will Achieve

In resolving this issue, we will achieve something far greater:

1. **A new language:** **Non-parametric structural equation models (SEMs)** — a richer, mechanistic foundation for the potential outcomes framework.
2. **A visual representation:** **Directed acyclic graphs (DAGs)** — a perfect fit for human-interpretable assumption elicitation.
3. **Interventions formalized:** A formal mathematical concept of an "intervention" that gives potential outcomes an **interventional interpretation**.
4. **Powerful tools:** The **Single World Intervention Graph (SWIG)** to deduce conditional exogeneity and to identify good vs. bad controls.

This framework is not confined to conditional exogeneity — it will help us derive other identification strategies later on.

Part I: From Domain Knowledge to Causal Graphs

Building the Influence Graph from an Empirical Example

The Hospital Messaging Example — Setup

Let's revisit the hospital messaging application from the in-class activity in the last lecture.

Setting

Doctors monitor young patients with diabetes through a remote monitoring system and send messages to incentivize better blood glucose control.

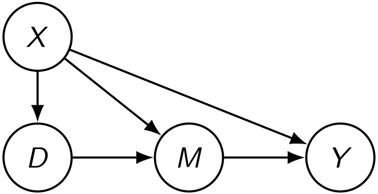
The key variables:

- Y = glucose time-in-range in the days after the message
- D = whether the doctor decided to send a message
- M = the actual text of the message
- X = all variables the doctor observes in the system (e.g., CGM data)

The Influence Graph — Observed Variables

Here is one reasonable "influence graph" for the observed variables:

- $X \to D$: CGM data influences the doctor's decision.
- $D \to M$: Decision to send leads to message text.
- $X \to M$: CGM data influences message content.
- $X \to Y$: Patient history influences outcome.
- $M \to Y$: The message influences glucose outcome.



But are there other variables that influence the observed variables?

Adding Unobserved / Latent Variables

Yes! There are unobserved variables that influence the observed ones:

Patient behavioral habits (U_p):

- What they eat, exercise habits, etc.
- $U_p \rightarrow X$: habits influence monitoring data
- $U_p \rightarrow Y$: habits influence glucose outcome

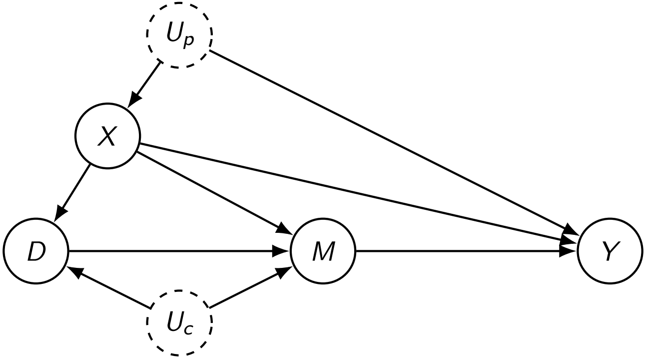
Doctor's capacity / workload (U_c):

- $U_c \rightarrow D$: workload affects decision to send
- $U_c \rightarrow M$: workload affects message quality

Noise terms ($\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$):

Each only influences its own node. We **omit these from the graph** for clarity and always keep in mind that each node has an associated independent noise term.

The Complete DAG



This is the **human-interpretable assumption** we are willing to make. This graph describes how we believe, after conversing with a domain expert, the data we observe are generated.

Dashed circles = unobserved. Each node also has implicit independent noise term ε_i.

The DAG as a Bayesian Network

This graph is a **Bayesian network** for the data:

Markov Factorization

The probability law of the data factorizes according to the graph:

$$p(\mathbf{v}) = \prod_i p\left(v_i \mid v_{\mathrm{Parents}(i)}\right)$$

What does this mean?

- Each variable is conditionally independent of its non-descendants given its parents.
- The graph captures **all conditional independencies** that hold in the data.

In our example:

$$p(u_p, u_c, x, d, m, y) = p(u_p)\, p(u_c)\, p(x \mid u_p)\, p(d \mid x, u_c)\, p(m \mid d, x, u_c)\, p(y \mid m, x, u_p)$$

Part II: Non-Parametric Structural Equations

What the DAG Really Means Mathematically

From DAG to Structural Equations

The DAG means: every variable is generated as a **function of its parent variables** and an **exogenous noise term**:

Non-Parametric Structural Equation Model (SEM)

$$
\begin{aligned}
X &= f_X(U_p, \varepsilon_X), \\
D &= f_D(X, U_c, \varepsilon_D), \\
M &= f_M(D, X, U_c, \varepsilon_M), \\
Y &= f_Y(M, X, U_p, \varepsilon_Y)
\end{aligned}
$$

- **No restriction** on how these functions look like (non-parametric!).
- The **crucial restriction** is on *what inputs* each function takes — only the **parents** of the node in the graph.
- Noise terms $\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ and latent factors U_p, U_c are **jointly independent**.

The Data Generating Process

The SEM defines a **sequential data generating process**:

1. **Draw** the jointly independent exogenous variables:
   $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y \sim$ some exogenous distribution.
2. **Generate** $X \leftarrow f_X(U_p, \varepsilon_X)$
3. **Generate** $D \leftarrow f_D(X, U_c, \varepsilon_D)$
4. **Generate** $M \leftarrow f_M(D, X, U_c, \varepsilon_M)$
5. **Generate** $Y \leftarrow f_Y(M, X, U_p, \varepsilon_Y)$

Key Insight

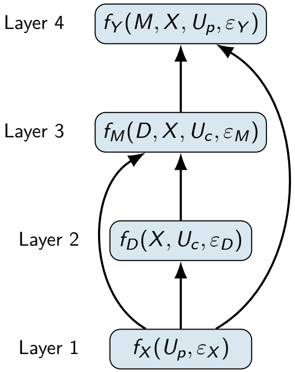
The DAG that we depicted from domain assumptions implicitly defines a **generative model** for the data!

Side Note: Connection to Deep Generative Models

There is an interesting new line of work that takes this viewpoint and connects it to **deep generative models**:

- Each structural equation f_i can be modeled as a **neural network layer**.
- The SEM becomes a deep generative model.
- Causal inference reduces to training deep generative models with gradient descent!

See: *Xia et al. (2021). "The Causal-Neural Connection."*



Poll 1: Understanding the SEM

Poll Question

In the hospital messaging SEM, which of the following is the **crucial restriction** imposed by the DAG?

(a) The structural equation functions f_i must be linear.

(b) Each function f_i can only take the **parents** of node i as inputs.

(c) The noise terms ε_i must be Gaussian.

(d) The latent factors U_p, U_c must be correlated.

Part III: Interventions & Potential Outcomes

What Does “Effect of Sending a Message” Really Mean?

What Is the "Effect of Sending a Message"?

Suppose we want to measure the **effect of sending a message** (any message).

In the language of potential outcomes:

$$Y(1) = \text{ outcome had we } \textit{sent} \text{ a message} \qquad Y(0) = \text{ outcome had we } \textit{not}.$$

But isn't this a bit too abstract? **How do we map this to the DGP we just described?**

A New Interpretation

Potential outcomes are the results of **interventions**: suppose we *intervened* and **forced** physicians to send a message (or not). Then:

- $Y(1)$: the **interventional outcome** when we force $D = 1$
- $Y(0)$: the **interventional outcome** when we force $D = 0$

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

Formalizing Interventions in the SEM

What does it **formally mean** to intervene? We go to the treatment variable D and instead of its natural value, we **fix** it to some value $d \in \{0, 1\}$.

DGP Under Intervention $\text{fix}(D = d)$

1. Draw $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ from **same distribution** as the factual world.
2. Generate $X \leftarrow f_X(U_p, \varepsilon_X)$ *(unchanged)*
3. Generate $D \leftarrow f_D(X, U_c, \varepsilon_D)$ *(natural value, still generated)*
4. But use the **fixed** value d downstream:
5. Generate $M(d) \leftarrow f_M(\mathbf{d}, X, U_c, \varepsilon_M)$
6. Generate $Y(d) \leftarrow f_Y(M(d), X, U_p, \varepsilon_Y)$

Invariance: The things that remain invariant across interventions are the distribution of the noise terms/latent factors and the structural equations.

Key Insights from the Interventional DGP

1. **Formal definition of POs:** We have given a formal mathematical definition of $Y(1)$ and $Y(0)$ using the *exact same building blocks* as the observed world.
2. **Consistency property:** If we set $d = D$, we recover the factual world:

$$Y = Y(D)$$

Bottom Line

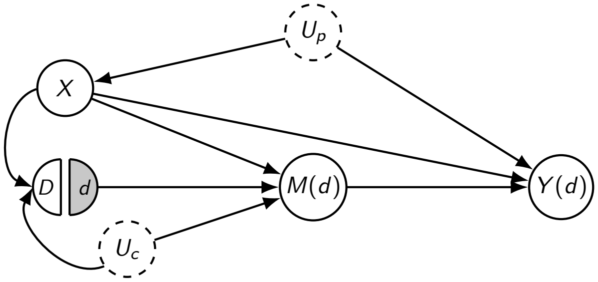
We have given an **interventional foundation** to the potential outcomes!

The Single World Intervention Graph (SWIG)

The interventional world under $\mathrm{fix}(D = d)$ can be depicted by a new Bayesian network:

SWIG Construction: "Strike Lightning on Node D"

Split D into two halves: **(i)** left semi-circle D (original) receives all **incoming** edges; **(ii)** right semi-circle d (fixed, shaded) sends all **outgoing** edges. Every variable **downstream** of the fixed value becomes an interventional random variable.



Checking Conditional Exogeneity on the SWIG

The SWIG is a **Bayesian network** for the interventional world. It contains both:

- D — the natural treatment that *would have been* assigned
- $Y(d)$ — the counterfactual outcome

So we can check **conditional exogeneity** by checking conditional independence in this Bayesian network!

What we need to check

In the SWIG, does conditioning on X block all paths between D and $Y(d)$?

$$Y(d) \perp\!\!\!\perp D \mid X \quad \textbf{???}$$

But first — how do we check conditional independence in a Bayesian network?

d-Separation: The Graphical Independence Criterion

Definition: d-Separation

Two nodes A and B are **d-separated** given a set $\mathbf{S}$ if **every** undirected path between A and B is **blocked**. A path is blocked if it contains:

1. A **chain** $\cdots \to W \to \cdots$ or **fork** $\cdots \leftarrow W \to \cdots$ where $W \in \mathbf{S}$, **or**
2. A **collider** $\cdots \to W \leftarrow \cdots$ where $W \notin \mathbf{S}$ and no descendant of W is in $\mathbf{S}$.

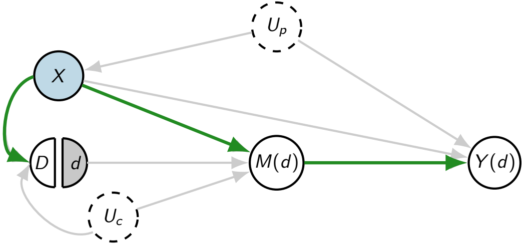
Key Property

In a Bayesian network, if A and B are d-separated given $\mathbf{S}$, then $A \perp\!\!\!\perp B \mid \mathbf{S}$.

Strategy: To check $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG, enumerate all paths between D and $Y(d)$ and check whether each is blocked by conditioning on X.

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

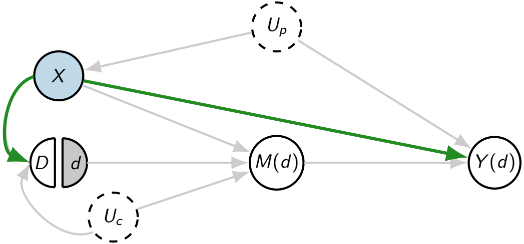
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:



Path 1: $D \leftarrow X \rightarrow M(d) \rightarrow Y(d)$ **Blocked** — fork at $X \in \mathbf{S}$

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

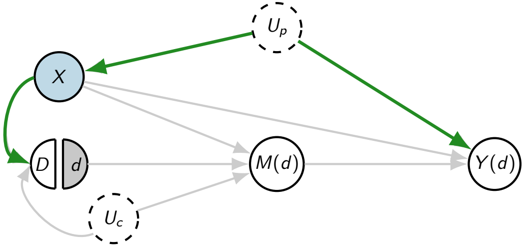
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:



Path 2: $D \leftarrow X \rightarrow Y(d)$ **Blocked** — fork at $X \in \mathbf{S}$

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

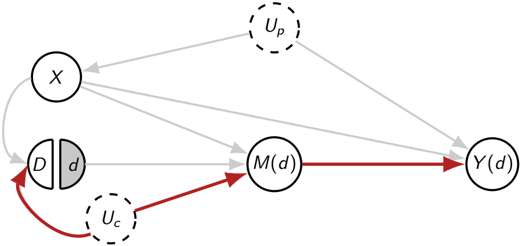
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:



Path 3: $D \leftarrow X \leftarrow U_p \rightarrow Y(d)$ **Blocked** — chain at $X \in \mathbf{S}$

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

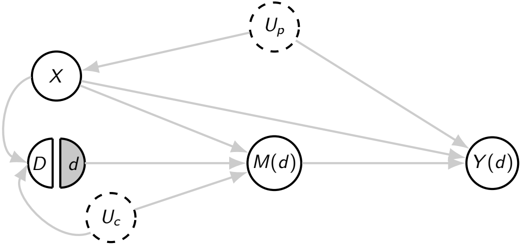
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:



Path 4: $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$ **Open!** — U_c unobserved, X not on path

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:



Conclusion

$Y(d) \not\perp\!\!\!\perp D \mid X$. Conditional exogeneity **fails**! The open path $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$ cannot be blocked because U_c is **unobserved**.

Why Does This Fail? — The Intuition

The Subtle Confounding Through the Mediator

The SWIG reveals a subtle but critical issue:

- U_c (doctor's workload/capacity) affects **both** the decision D **and** the message quality M.
- When we only intervene on D (force everyone to send a message), the **quality** of $M(d)$ is still influenced by U_c.
- This creates a backdoor path: $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$.

In plain language:

- In the observed data, doctors who *chose* to send messages typically had **more time available**, so they sent **better, more effective** messages.
- If we force *all* doctors to send messages, time-limited doctors would send **poor quality** messages.
- The observed efficacy of "sending a message" **overestimates** the true causal effect of $\text{fix}(D = 1)$.

A Better Intervention: $\text{fix}(M = m)$

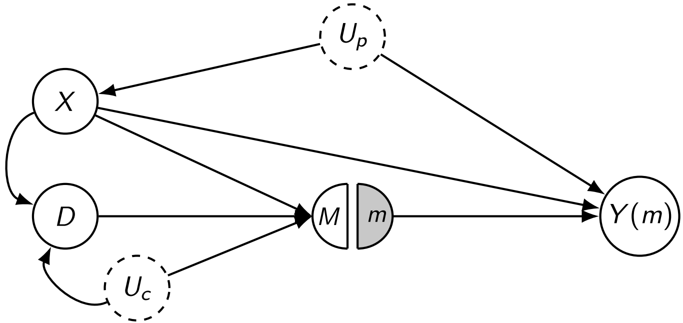
Since the problem is confounding through the mediator, what if we intervene on the **message itself**?

DGP Under $\text{fix}(M = m)$

1. Draw $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ (same distribution).
2. $X \leftarrow f_X(U_p, \varepsilon_X)$
3. $D \leftarrow f_D(X, U_c, \varepsilon_D)$ (natural value)
4. $M \leftarrow f_M(D, X, U_c, \varepsilon_M)$ (natural value, still generated)
5. But use the **fixed message** m downstream:
6. $Y(m) \leftarrow f_Y(\mathbf{m}, X, U_p, \varepsilon_Y)$

Key: We still generate D and M naturally (we need them in the SWIG), but the outcome equation uses the **fixed** message m.

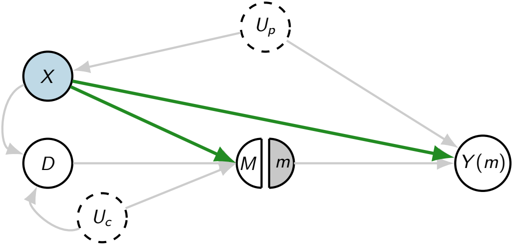
SWIG for $\mathrm{fix}(M = m)$



Now D is **not** split — it is a regular node. Only M is intervened upon. The outcome $Y(m)$ depends on the fixed message m, not on the natural M.

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

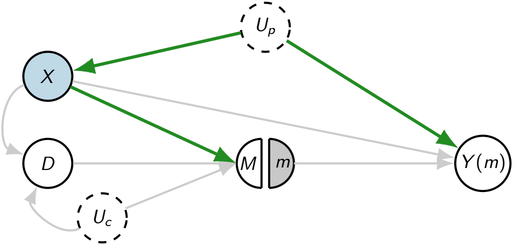
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



Path 1: $M \leftarrow X \rightarrow Y(m)$ **Blocked** — fork at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

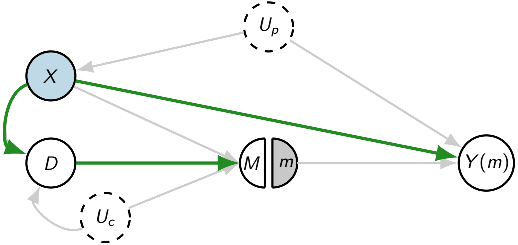
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



Path 2: $M \leftarrow X \leftarrow U_p \rightarrow Y(m)$ **Blocked** — chain at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

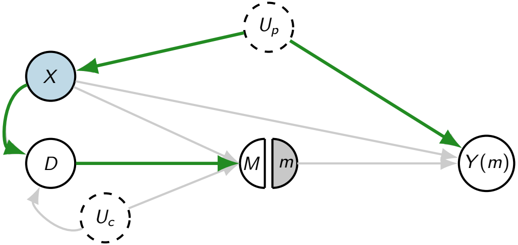
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



Path 3: $M \leftarrow D \leftarrow X \rightarrow Y(m)$ **Blocked** — chain at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

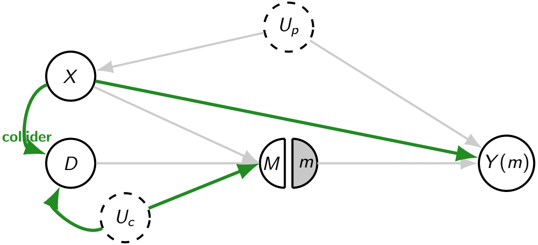
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



Path 4: $M \leftarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$ **Blocked** — chain at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

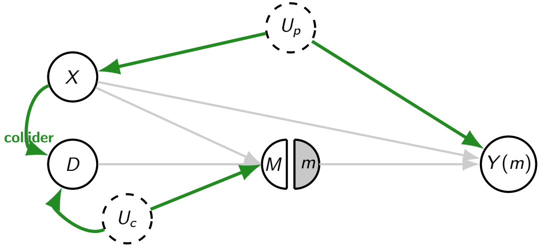
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



Path 5: $M \leftarrow U_c \rightarrow D \leftarrow X \rightarrow Y(m)$ **Blocked** — collider at D, $D \notin \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

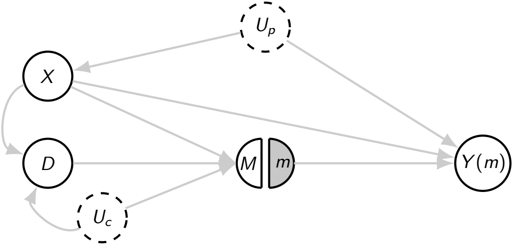
Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



Path 6: $M \leftarrow U_c \rightarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$ **Blocked** — collider at D

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



All Paths Blocked!

Every path from M to $Y(m)$ is blocked by X. Paths through U_c hit the **collider** at D ($U_c \rightarrow D \leftarrow X$), which blocks them without conditioning.

Conditional Exogeneity **Holds** for $\text{fix}(M = m)$!

Result

$$Y(m) \perp\!\!\!\perp M \mid X \quad \checkmark$$

Conditional exogeneity holds when we intervene on the **message** M. We can apply all identification and estimation strategies from prior lectures to estimate the causal effect of the message content!

Voilà! We have deduced conditional exogeneity as a **conclusion**, where the only assumption was the **human-interpretable DAG**.

But note: we could **not** *have deduced this for the binary decision D alone — the SWIG revealed the subtle confounding through the mediator!*

Intervening on Multiple Variables

We can also intervene on **both** D and M simultaneously:

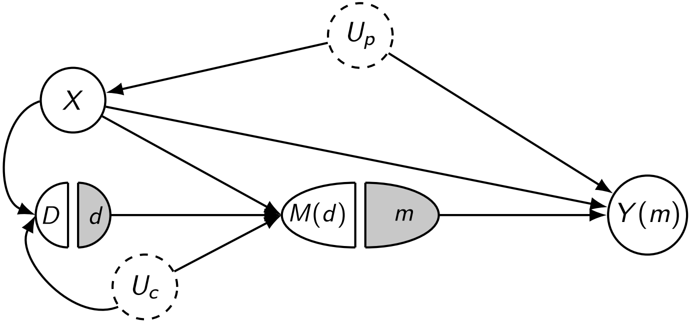
DGP Under $\text{fix}(D = d, M = m)$

1. Draw $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ (same distribution).
2. $X \leftarrow f_X(U_p, \varepsilon_X)$
3. $D \leftarrow f_D(X, U_c, \varepsilon_D)$ (natural value)
4. $M(d) \leftarrow f_M(d, X, U_c, \varepsilon_M)$ (use fixed d)
5. But instead of $M(d)$, use the **fixed message** m:
6. $Y(d, m) \leftarrow f_Y(m, X, U_p, \varepsilon_Y)$

Notice: $Y(d, m) = Y(m)$ since d doesn't directly enter f_Y.

SWIG for Double Intervention

Now we are "striking two lightnings" — intervening on both D and M:



We cannot check the condition $Y(m) \perp\!\!\!\perp (D, M) \mid X$ on this SWIG!

Note: On this SWIG, M does not appear as a standalone variable — only $M(d)$ (left half) and m (right half).

If you cannot verify it on the SWIG you should not use it for identification!

What We've Achieved So Far

Through the hospital messaging example, we have:

1. **Started** from interpretable domain assumptions $\to$ **DAG**
2. **Formalized** what the DAG means mathematically $\to$ **Non-Parametric SEM**
3. **Defined** what an intervention is and the DGP under intervention
4. **Given** an interventional interpretation of potential outcomes and ATE
5. **Used** the SWIG to check conditional exogeneity — and discovered it **fails** for $\text{fix}(D = d)$ but **holds** for $\text{fix}(M = m)$!

The **only assumption** was the human-interpretable DAG. Conditional exogeneity was a **conclusion** to check, not a premise — and the SWIG revealed a subtle failure we might have missed!

The Power of the SWIG: Avoiding Subtle Mistakes

The Mistake We Almost Made

Without the SWIG, we might have naively assumed $Y(d) \perp\!\!\!\perp D \mid X$ and estimated the effect of "sending a message" (D) by conditioning on patient data (X). This would have been **wrong**.

What the SWIG revealed:

- There is **implicit confounding through the mediator** M: the doctor's workload (U_c) affects both the *decision* to send and the *quality* of the message.
- In the observed data, doctors who chose to send messages typically had **more time available** $\Rightarrow$ they sent **better, more effective** messages.
- If we force *all* doctors to send messages, time-limited doctors would send **poor quality** messages — the observed efficacy **overestimates** the true effect.

The fix: Intervene on M directly, or additionally condition on physician workload (U_c).

Technical Parenthesis: Primitive Potential Outcomes

We can push interventions to the extreme — fix **all** variables, even the latent factors:

Primitive Interventional Variables

$$
\begin{aligned}
X(u_p) &\leftarrow f_X(u_p, \varepsilon_X) \\
D(x, u_c) &\leftarrow f_D(x, u_c, \varepsilon_D) \\
M(d, x, u_c) &\leftarrow f_M(d, x, u_c, \varepsilon_M) \\
Y(m, x, u_p) &\leftarrow f_Y(m, x, u_p, \varepsilon_Y)
\end{aligned}
$$

These are the **primitive potential outcomes** — the counterfactual outcome of a variable when we fix *all* of its parent variables.

Because the noise terms are independent, these primitive POs are **jointly independent**.

Alternative Definition of Structural Equations: POs as Primitives

An equivalent way to define the whole framework:

Step 1: Nature generates all primitive POs from some exogenous distribution satisfying joint independence:

$$\{Y(m,x,u_p)\},\ \{M(d,x,u_c)\},\ \{D(x,u_c)\},\ \{X(u_p)\},\ U_p,\ U_c$$

Step 2: Then generate the observed data by successive "plugging in":

$$X \leftarrow X(U_p), \quad D \leftarrow D(X,U_c), \quad M \leftarrow M(D,X,U_c), \quad Y \leftarrow Y(M,X,U_p)$$

Mathematical Equivalence Between PO and SEMs

The Two Sides of the Same Coin

The SEM framework and the potential outcomes framework are equivalent formulations. The SEM gives a **mechanistic** foundation of how the primitive counterfactuals are generated; the PO framework gives a **counterfactual** foundation.

Two Meeting Points

Typically, we care about average **interventional counterfactuals**, not averages of these primitive potential outcomes. The interventional counterfactuals are the most natural analogue of what people typically define as the potential outcomes in the potential outcomes framework.

Takeaways

1. **DAGs as Domain Assumptions:** A directed acyclic graph encodes human-interpretable assumptions about the data generating process. The associated **non-parametric SEM** formalizes it mathematically.
2. **Interventional Potential Outcomes:** Potential outcomes arise naturally as **interventional random variables** from the SEM. An intervention means "fixing" a variable to a value in the structural equations.
3. **SWIGs Reveal Subtle Confounding:** The SWIG showed that $Y(d) \not\perp\!\!\!\perp D \mid X$ due to implicit confounding through the mediator, but $Y(m) \perp\!\!\!\perp M \mid X$ holds. Without the SWIG, this subtle failure is easy to miss.
4. **Choose the Right Intervention:** The causal question matters: intervening on the **message content** M is identifiable; intervening on the **binary decision** D alone is not (without controlling for workload).

References

- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge University Press.
- Richardson, T. S., & Robins, J. M. (2013). Single World Intervention Graphs (SWIGs). Working Paper, University of Washington.
- Chernozhukov, V., Hansen, C., Kallus, N., Spindler, M., & Syrgkanis, V. (2026). *Applied Causal Inference Powered by ML and AI*. §7, §11.
- Xia, K., Lee, K., Bengio, Y., & Bareinboim, E. (2021). The Causal-Neural Connection. *NeurIPS 2021*.
- Cinelli, C., Forney, A., & Pearl, J. (2022). A Crash Course in Good and Bad Controls. *Sociological Methods & Research*.